A
Project Report
On
Medilog Portal

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CHARUSAT
CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CERTIFICATE

This is to certify that the report entitled "**Medilog**" is a bonafied work carried out by **Heer Modi (22DIT036)** under the guidance and supervision of Asst. Prof. Pooja Singh Chaudhary for the subject **Minor Project (IT452)** of 7th Semester of Bachelor of Technology in **Information Technology** at Devang Patel Institute of Advance Technology and Research (DEPSTAR), Faculty of Technology & Engineering (FTE) – CHARUSAT, Gujarat.

To the best of my knowledge and belief, this work embodies the work of candidate himself, has duly been completed, and fulfills the requirement of the ordinance relating to the B.Tech. Degree of the University and is up to the standard in respect of content, presentation and language for being referred to the examiner.

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I hereby declare that the project report entitled "**Medilog**" submitted by me to Devang Patel Institute of Advance Technology and Research (DEPSTAR), Changa in partial fulfilment of the requirements for the award of the degree of **B.Tech Information Technology**, from, DEPSTAR, FTE is a record of bonafide IT452 Minor Project carried out by me under the guidance of **Pooja Singh Chaudhray**. I further declare that the work carried out and documented in this project report has not been submitted anywhere else either in part or in full and it is the original work, for the award of any other degree or diploma in this institute or any other institute or university.

Heer Modi(22DIT036)

This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

ABSTRACT

The project titled “MediLog Portal” focuses on the development of a comprehensive digital healthcare management system using the MERN stack (MongoDB, Express.js, React.js, and Node.js) integrated with Cloudinary for secure and efficient file storage. The portal is designed to streamline the process of storing, managing, and sharing medical records by providing a centralized, role-based platform for patients and healthcare professionals. MediLog modernizes traditional record handling by offering a paperless, accessible, and user-friendly solution that enhances communication between doctors and patients while ensuring the confidentiality of medical data.

The backend, built using Express.js and Node.js, handles authentication, user management, and data routing through a secure and scalable API structure. MongoDB serves as the primary database for storing patient profiles, medical histories, and uploaded records with flexibility and reliability. The frontend, developed with React.js and Material UI (MUI), delivers a clean, modern, and responsive interface, ensuring an intuitive user experience across all devices. Integration with Cloudinary provides encrypted cloud storage for medical reports, prescriptions, and diagnostic images, maintaining data security and ease of access.

MediLog offers key features such as secure user authentication, medical record uploads, role-based dashboards, and sharing capabilities between patients and doctors. Additionally, it includes intelligent components like reminder tracking and progress monitoring, which can be further enhanced with AI-driven insights in future versions.

Overall, MediLog Portal demonstrates how technology can simplify healthcare data management while promoting accessibility, privacy, and efficiency. It establishes a strong foundation for future expansion toward AI integration, wearable device connectivity, and advanced analytics to further empower both patients and healthcare professionals in a connected digital ecosystem.

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to everyone who supported and guided us throughout the development of our project, **"MediLog Portal."** This project has been an excellent opportunity to translate theoretical learning into practical experience, strengthening our understanding of full-stack web development and modern software design. The journey of developing MediLog has helped us grow both technically and personally, teaching us the importance of collaboration, problem-solving, and persistence.

We extend our heartfelt thanks to our respected guide, **Mr. Ashish Katira**, Assistant Professor, **Department of Information Technology, DEPSTAR-FTE, CHARUSAT**, for his continuous encouragement, expert advice, and valuable feedback. His mentorship and technical insights were crucial in shaping the project's architecture, improving its functionality, and maintaining our focus throughout the process. We are also grateful to the **Department of Information Technology** for providing us with the necessary infrastructure, resources, and supportive environment that made this work possible.

We would also like to acknowledge the help and cooperation of our **friends and peers**, who provided constructive suggestions during various stages of development and testing. Lastly, we extend our deepest gratitude to our **families** for their constant motivation, patience, and moral support, which inspired us to complete this project with dedication and enthusiasm.

With gratitude,
Heer Modi – 22DIT036

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CHAPTER – 1
Project Definition

The project titled **"MediLog Portal"** aims to develop a modern healthcare information system that enables patients and doctors to efficiently manage and share medical records through a secure digital platform. The portal provides an organized and centralized system for storing medical documents such as prescriptions, test reports, and diagnostic results, promoting better communication between patients and healthcare providers. It replaces traditional paper-based methods with a more reliable and accessible digital solution that ensures data privacy, accuracy, and easy retrieval.

The system is designed with a **role-based structure**, where patients can upload, categorize, and share their records, while doctors can view patient histories, monitor progress, and offer digital consultations. This structured interaction helps improve the quality of healthcare services and supports informed medical decisions. The platform also emphasizes a user-friendly and responsive interface to ensure smooth navigation across all devices.

MediLog is developed using the **MERN Stack (MongoDB, Express.js, React.js, and Node.js)** to ensure flexibility, scalability, and fast performance. **Cloudinary** is integrated for secure and efficient file storage, while **Material UI** provides a clean and modern design for the frontend. The key modules include **Authentication**, **Record Management**, **Dashboard Interface**, **File Sharing**, and an **AI-based Reminder System** that assists patients in tracking medications and appointments. Overall, MediLog provides a secure, efficient, and convenient solution for managing healthcare records in the digital age.

CHAPTER – 2

Introduction

2.1 Overview

The **"MediLog Portal"** is a web-based healthcare management system developed using the **MERN Stack** designed to simplify and digitize the way patients and doctors manage medical records. The project focuses on developing an interactive, user-friendly, and secure platform that enables patients to upload, organize, and share their health data, while allowing doctors to review patient histories, provide consultations, and track progress seamlessly. MediLog bridges the communication gap between healthcare providers and patients by offering a centralized system where both parties can access and manage essential medical information conveniently and efficiently.

The portal aims to eliminate the drawbacks of traditional paper-based medical record systems, such as data loss, inefficiency, and inaccessibility, by introducing a digital solution that ensures data integrity and availability at all times. Built with scalability and performance in mind, MediLog utilizes **MongoDB** for storing patient profiles and medical data, ensuring flexibility in handling various file types and structures. **Express.js** and **Node.js** power the backend, managing authentication, secure API communication, and data flow between the client and the database. The **React.js** frontend, styled with **Material UI**, provides an intuitive and responsive interface that enhances usability and accessibility across multiple devices.

Key aspects of MediLog include secure **role-based authentication**, **medical record management**, **file sharing**, and **AI-powered smart reminders** that assist patients in staying on track with their treatments and appointments. The platform is developed to ensure data security, transparency, and efficiency, making it a valuable digital tool for healthcare institutions, clinics, and individual practitioners aiming to improve the accessibility and management of patient health information. By focusing on security, usability, and real-time accessibility, MediLog serves as a step forward in digital healthcare innovation and patient empowerment.

2.2 Purpose

The purpose of the **"MediLog Portal"** is to provide a unified digital platform that simplifies healthcare data management and improves communication between patients and medical professionals. The portal is designed to address the inefficiencies of traditional healthcare systems, where medical records are often scattered, difficult to access, and vulnerable to loss. MediLog offers a structured and secure system where patients can easily upload, categorize, and share medical documents such as prescriptions, lab results, and diagnostic images, ensuring that their complete medical history is always accessible and organized.

From the doctor's perspective, the system provides streamlined access to patient health data, allowing physicians to make informed decisions based on accurate and up-to-date medical records. Through its interactive dashboard and data-sharing capabilities, MediLog enhances collaboration and communication, ensuring that patients receive timely advice and treatment. The platform promotes digital transformation in healthcare by introducing automation and convenience into the record-keeping and consultation processes, ultimately saving time and reducing administrative burden for both parties.

Beyond basic record management, MediLog also focuses on enhancing patient engagement and responsibility toward their own health. Features like AI-based reminders for medication and appointments, data sharing for second opinions, and secure document storage collectively empower users to take an active role in their health journey. The project ultimately aims to contribute to the evolution of smart healthcare systems by providing a secure, efficient, and patient-centered digital solution that supports better healthcare delivery and improved medical outcomes.

2.3 Scope

The scope of the **"MediLog Portal"** project encompasses the complete design, development, and deployment of a full-stack healthcare management system that facilitates the digital handling of medical records and promotes seamless interaction between patients and doctors. The project focuses on building a reliable and scalable platform that ensures security, ease of use, and accessibility. Major functionalities and components of the system include:

- **User Authentication:** Implementing a secure, role-based login system using JWT and bcrypt.js to safeguard patient and doctor credentials, ensuring only authorized users can access sensitive medical data.
- **Profile Management:** Allowing patients and doctors to manage their personal details, contact information, and healthcare-related profiles efficiently.
- **Medical Record Upload and Management:** Enabling patients to upload documents such as prescriptions, reports, and diagnostic results to Cloudinary, ensuring secure cloud-based storage with organized folder categorization.

- **Record Sharing:** Providing patients with options to share specific medical documents with doctors for consultations, maintaining privacy and controlled access.
- **Doctor Dashboard:** Offering doctors an interface to view patient data, monitor treatment progress, and provide follow-ups through a structured and interactive dashboard.
- **AI-Based Reminder and Notification System:** Incorporating automated reminders to notify patients about upcoming checkups, medicine schedules, and other important updates.
- **Data Management and Security:** Utilizing **MongoDB** for structured and unstructured data storage, ensuring reliability, scalability, and data encryption for privacy compliance.
- **Responsive UI/UX:** Designing the front-end with **React.js and Material UI** to create a clean, modern, and responsive interface that functions smoothly across all devices.
- **Future Integration and Scalability:** Preparing the platform for future enhancements such as wearable health device connectivity, AI-based analytics, and hospital system integration to support large-scale healthcare operations.

The MediLog Portal aims to serve as a comprehensive healthcare management solution that addresses the needs of both patients and doctors, offering an efficient, secure, and intelligent system for maintaining medical records and improving healthcare accessibility through digital innovation.

CHAPTER – 3

Tools & Technologies Used with Justification

1. **MERN Stack**:
    - **MongoDB**: Stores patient profiles and medical records.
        - *Justification*: NoSQL allows dynamic, varied data structures, ideal for this project.
    - **React.js**: Builds responsive user interfaces for dashboards, appointment booking, and notifications.
        - *Justification*: React ensures seamless, dynamic user experiences with its component-based structure.
    - **Node.js**: Handles backend operations, user authentication, and data routing.
        - *Justification*: Node.js is efficient for handling real-time requests and data management.
    - **Express.js**: Manages API routes and server-side logic.
        - *Justification*: Simplifies backend with powerful routing and middleware.
2. **Authentication & Security**:
    - **JWT**: Manages secure user authentication and session management.
        - *Justification*: JWT ensures stateless, scalable authentication.
    - **BCrypt.js**: Hashes passwords securely.
        - *Justification*: Ensures password security, protecting user credentials.
3. **Communication & Notifications**:
    - **Socket.io**: Delivers real-time notifications for appointments and updates.
        - *Justification*: Provides instant, bi-directional communication for real-time engagement.
4. **Data Handling**:
    - **Multer**: Handles file uploads, like supporting documents for complaints.
        - *Justification*: Efficiently manages secure file uploads.
    - **Validator.js**: Validates and sanitizes user inputs.

        - *Justification*: Ensures data consistency and prevents malicious inputs.

5. **Programming Language**:
    - **JavaScript**: Powers both front-end and back-end development.
        - *Justification*: Its versatility supports full-stack development with the MERN stack.
6. **Dataset**:
    - **User and Counselor Data**: Manages profiles, appointments and feedback.
        - *Justification*: Efficient data handling

CHAPTER – 4

Hardware & Software Requirements

 Hardware **Requirements**:

- **Processor**: Intel i5 or above
- **RAM**: 8 GB minimum (16 GB recommended)
- **Storage**: 256 GB SSD

 Software **Requirements**:

- **Operating System**: Windows 10 or Ubuntu 20.04
- **Node.js** (for backend development)
- **MongoDB** (for database management)
- **React.js** (for frontend development)
- **Express.js** (for API handling)
- **Python 3.8 or above** (for additional scripting, if necessary)
- **Git** (for version control)
- **Postman** (for API testing)

CHAPTER – 5
Workflow diagram

- This chapter presents the workflow of the MediLog system through detailed diagrams. It includes the class diagram and flowchart that illustrate the system's structure and operational flow.

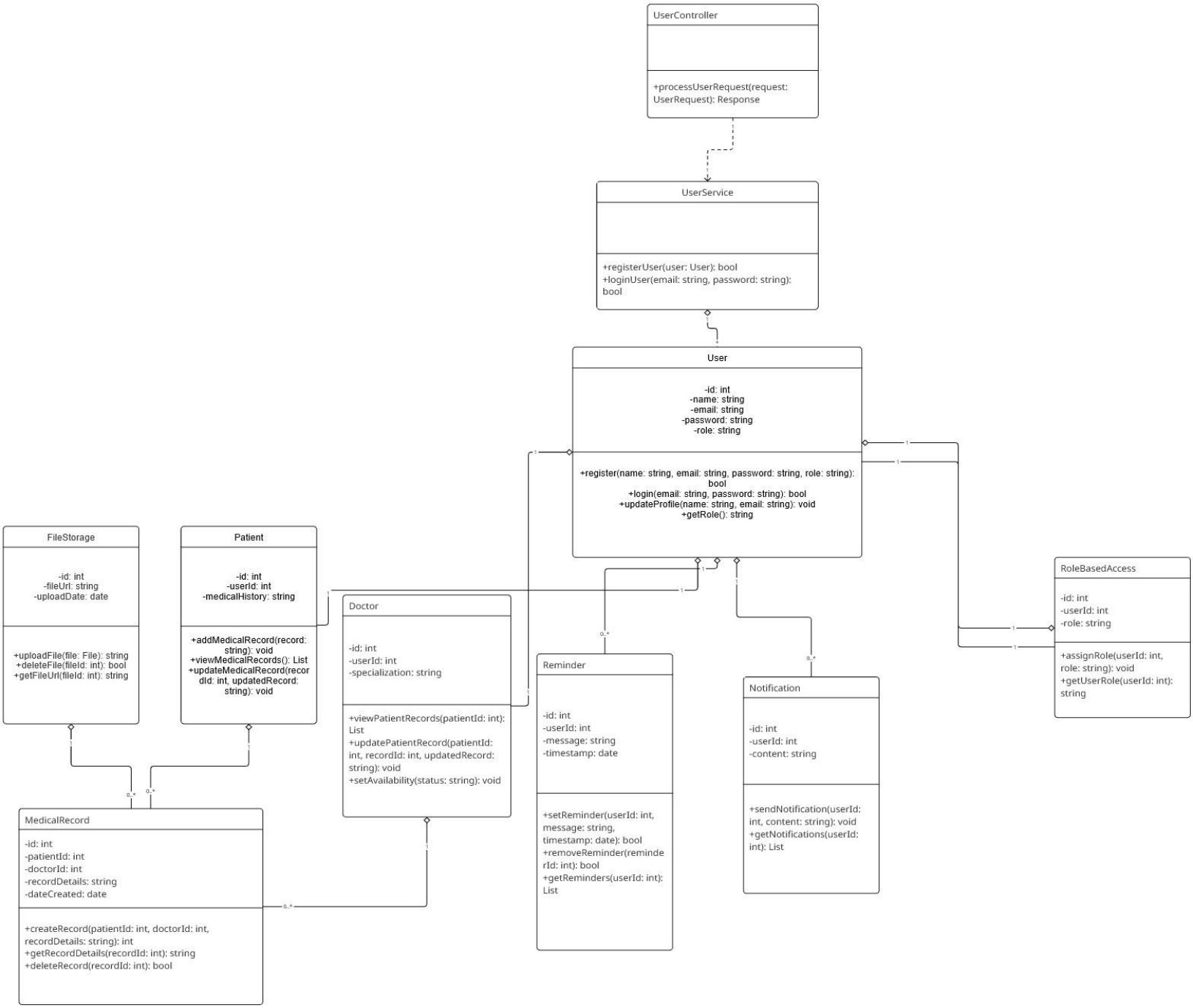


FIG 5. 1 CLASS DIAGRAM

The class diagram represents the structural design of the MediLog system, showing the relationships between different classes and their functionalities.

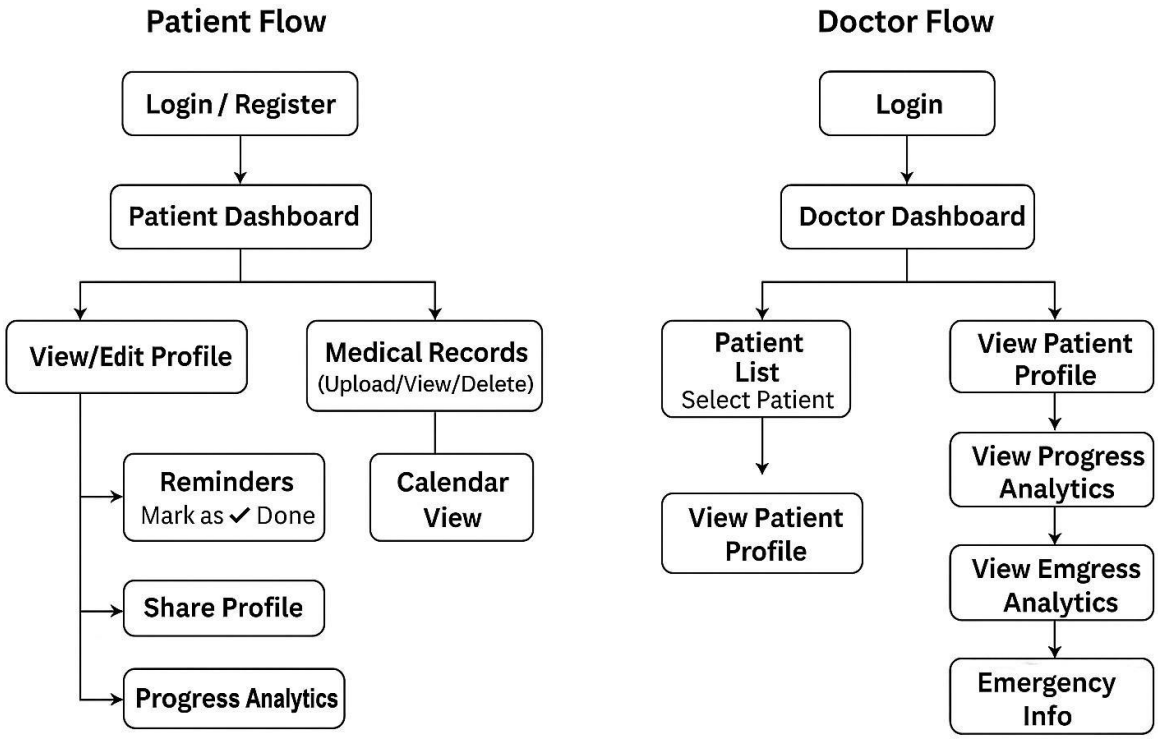


FIG 5. 2 FLOW CHART

The flowchart diagram illustrates the step-by-step process and logical flow of operations within the MediLog system.

CHAPTER – 6

Implementation screenshot

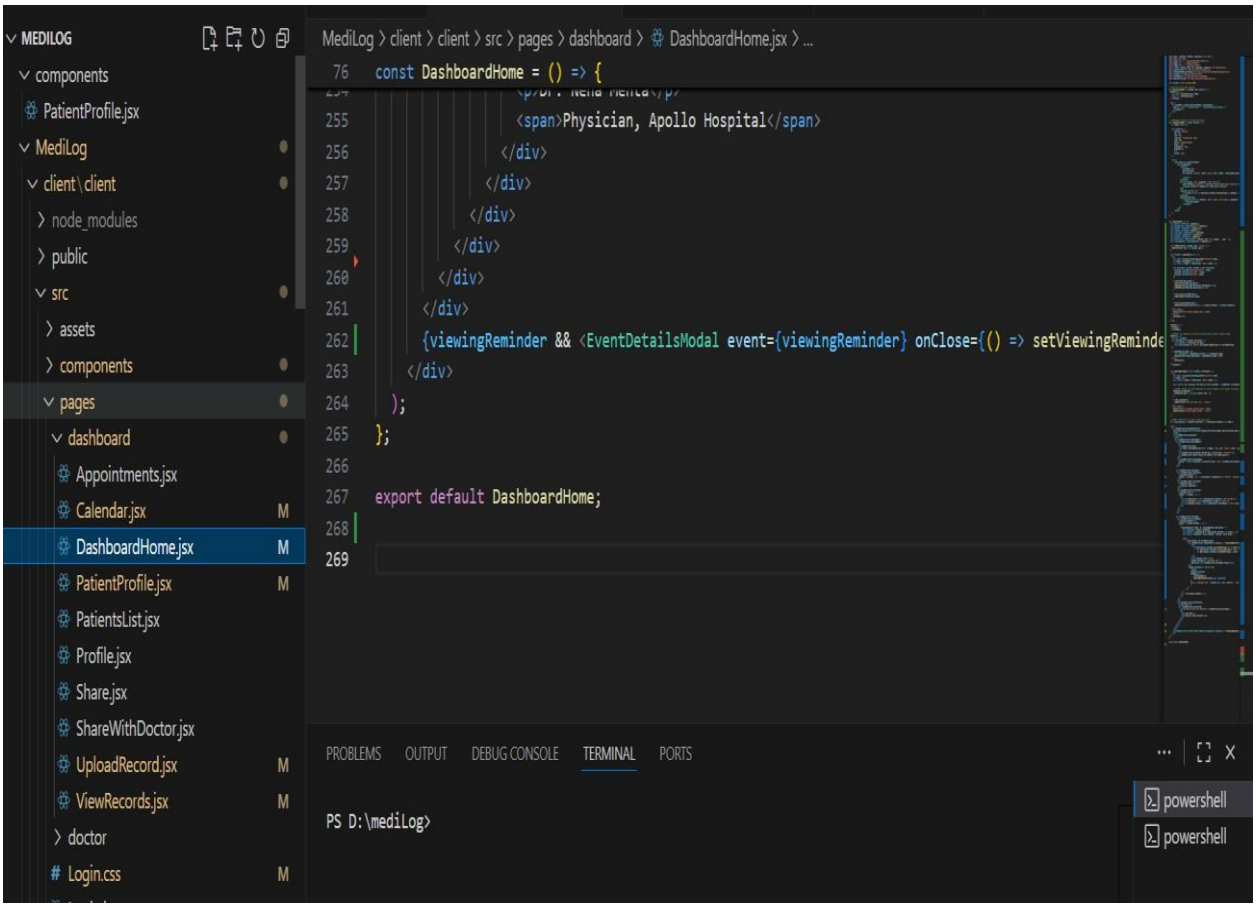


FIG 6.1 FRONTEND IMPLEMENTATION

The screenshot displays the frontend interface of the MediLog system, showcasing its user-friendly and visually engaging design.

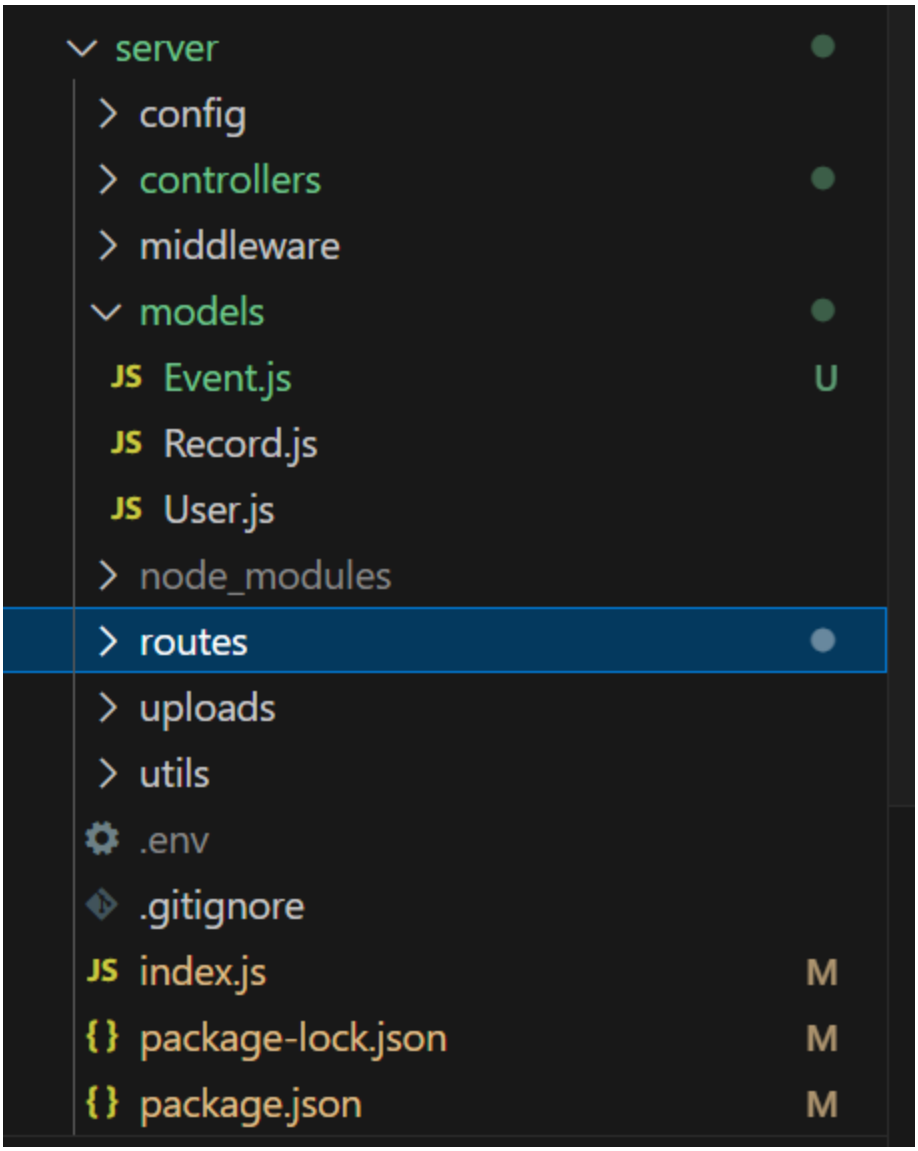


FIG 6. 2 BACKEND IMPLEMENTATION

The screenshot displays the backend code of the MediLog system, highlighting the server-side logic and API integration for efficient data handling.

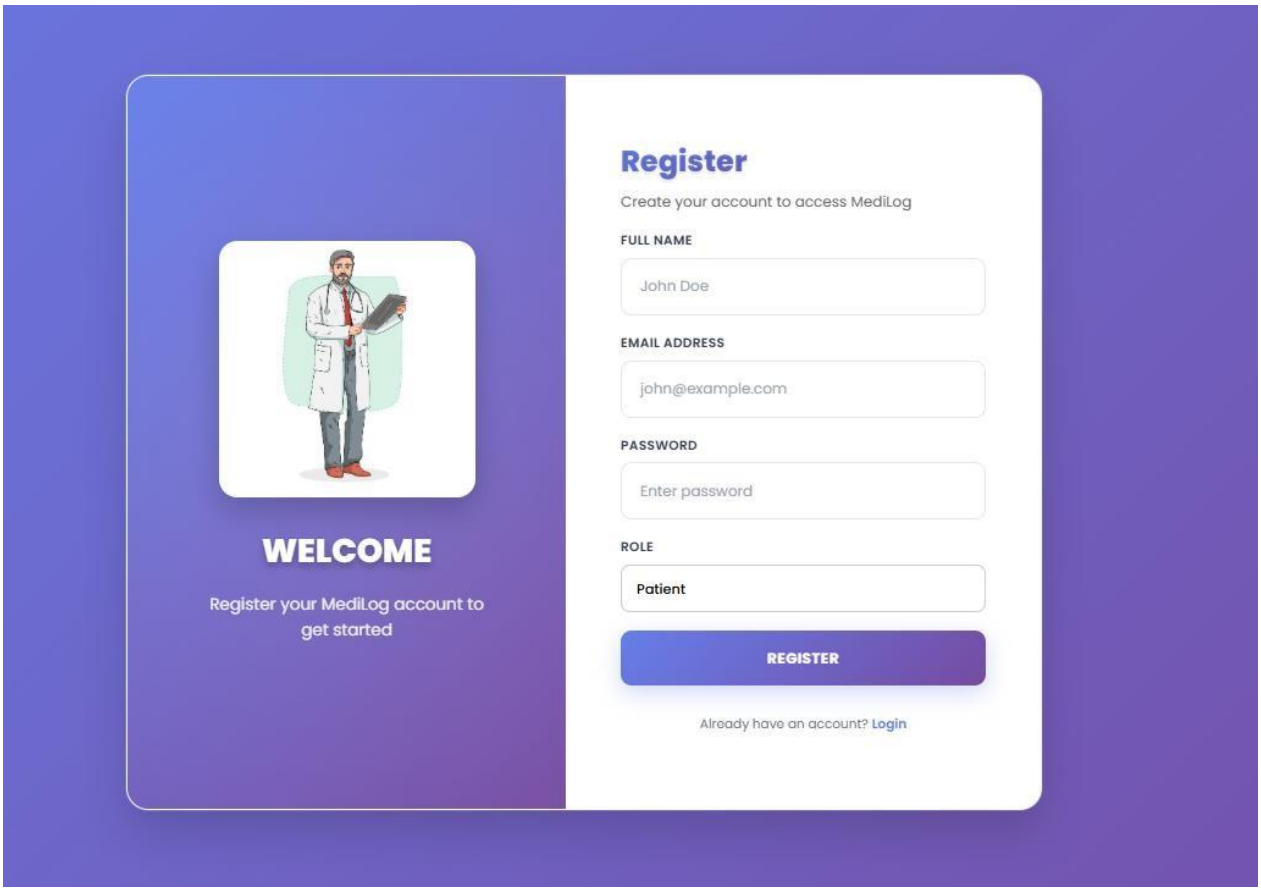


FIG 6. 3 REGISTRATION PAGE

The screenshot shows the registration page of the MediLog system, allowing new users to create an account by entering their details and selecting their role.

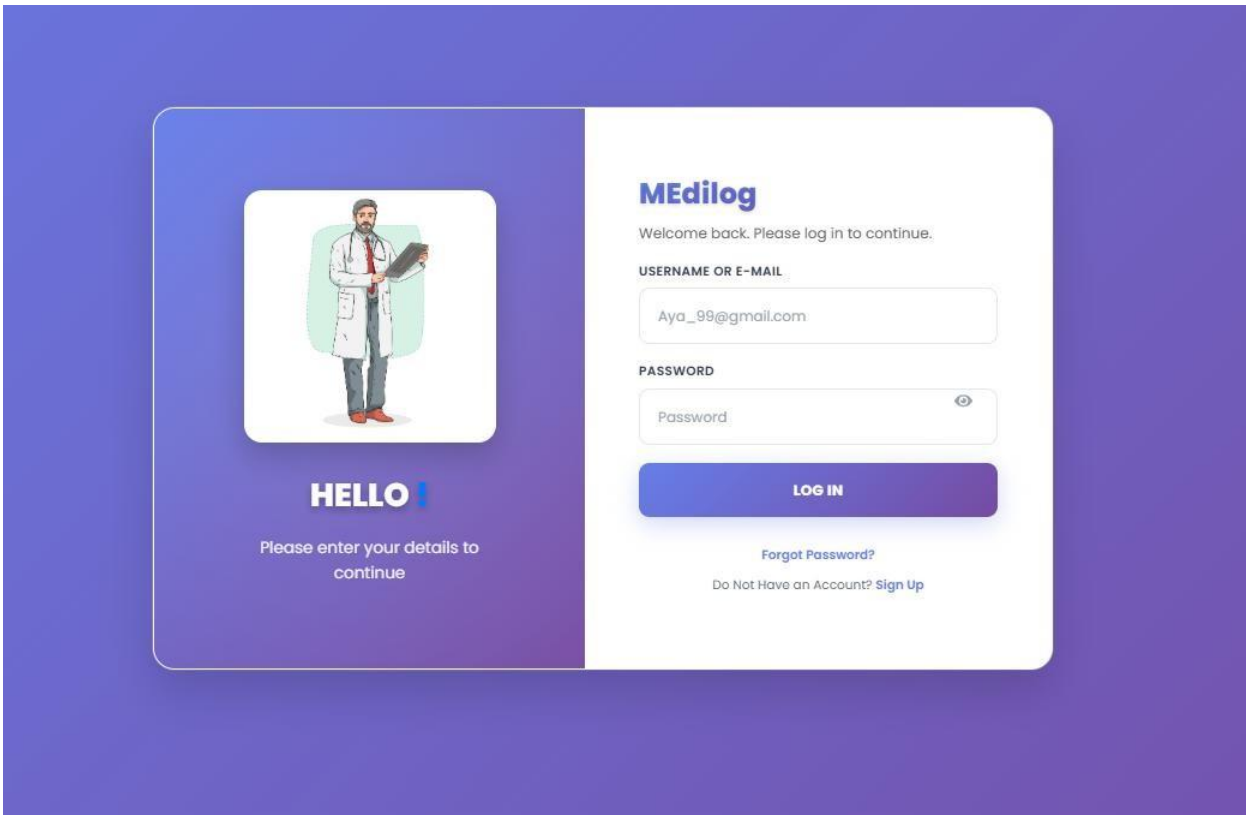


FIG 6. 4 LOGIN PAGE

The screenshot shows the role-based login page of the MediLog system, enabling secure access for patients and doctors based on their assigned roles.

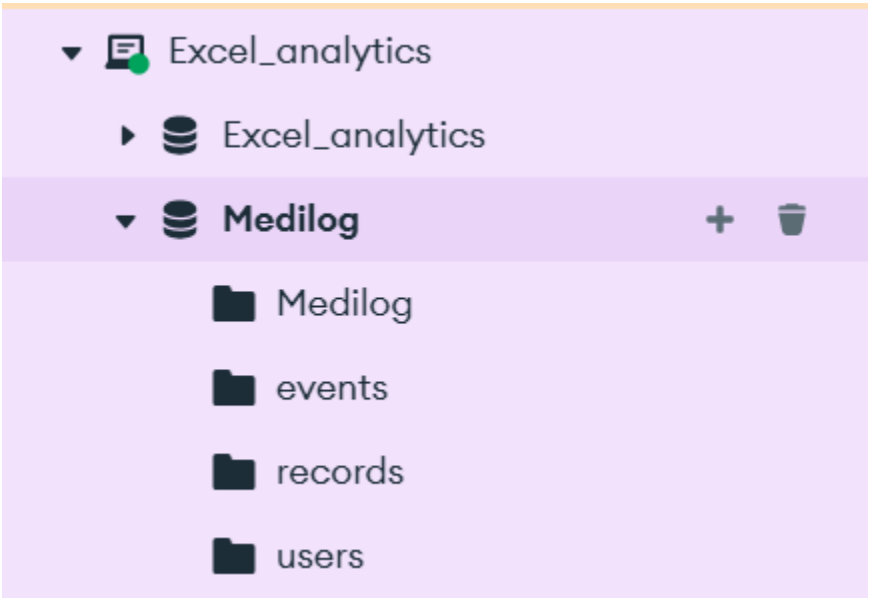


FIG 6. 5 DATABASES

The screenshot displays the MediLog system's database structure, showcasing how user and medical record data are stored and organized efficiently.

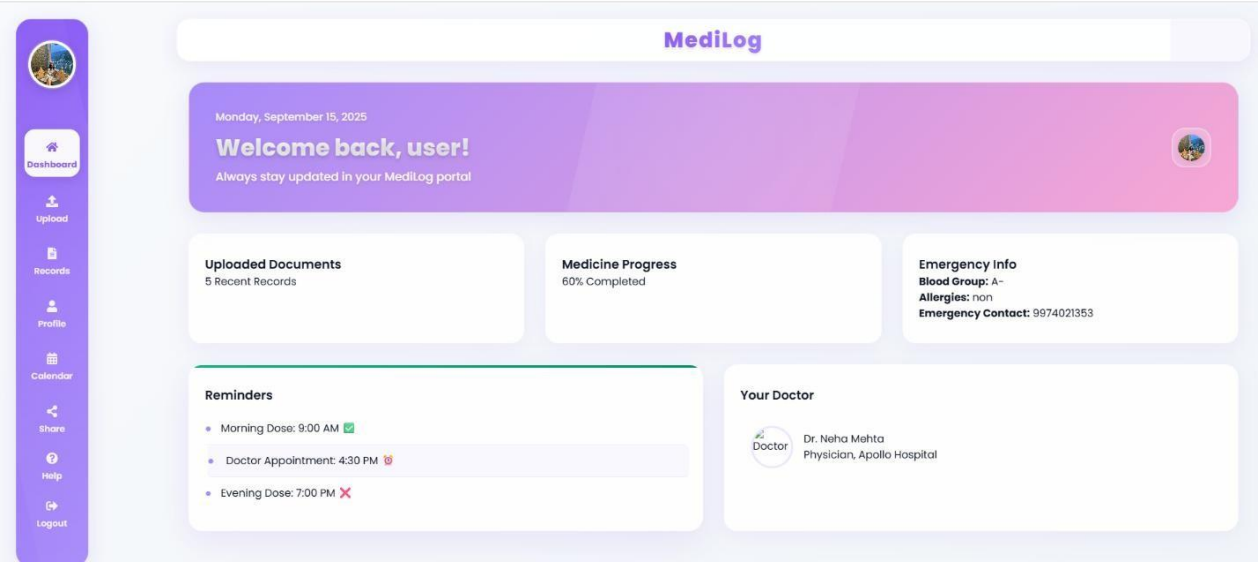


FIG 6. 6 PATIENT DASHBOARD PAGE

The screenshot presents the main patient dashboard, offering easy navigation and quick access to essential health features.

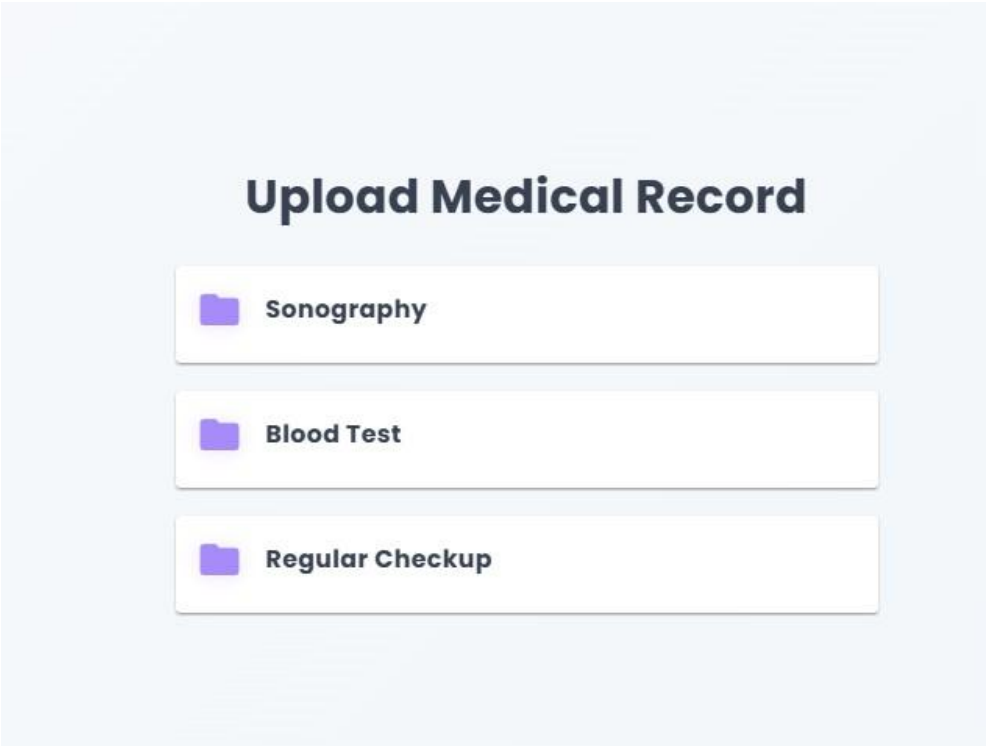


Fig 6.7 Upload Document

The screenshot shows the patient dashboard section where users can securely upload their medical documents to the MediLog system.

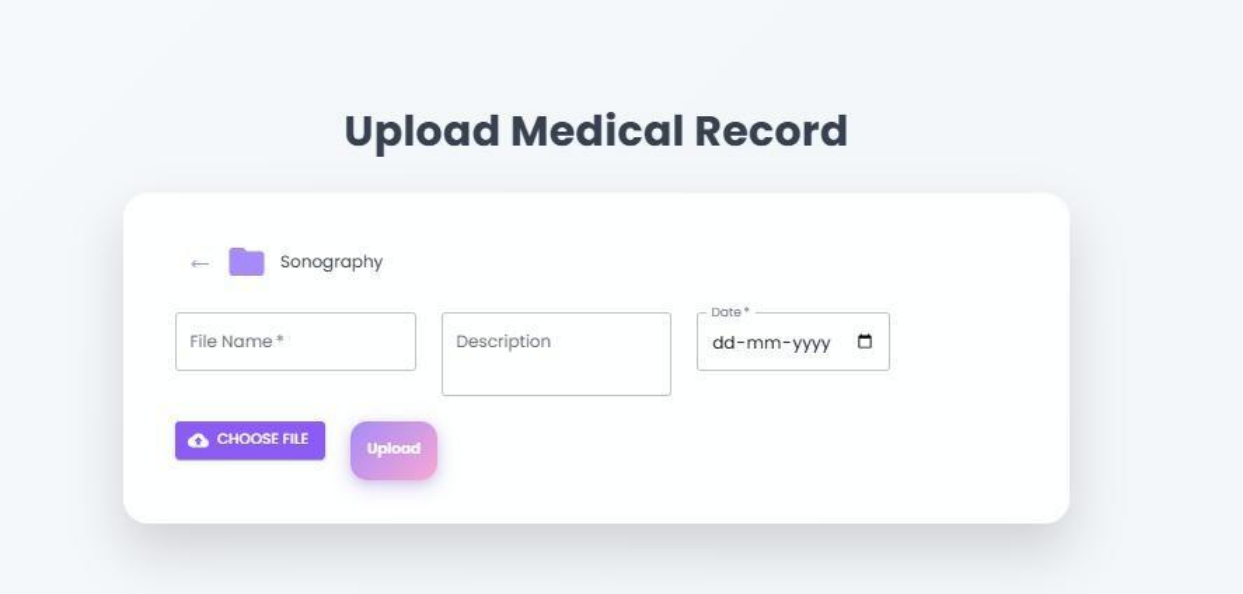


Fig 6.8 Uploading Form

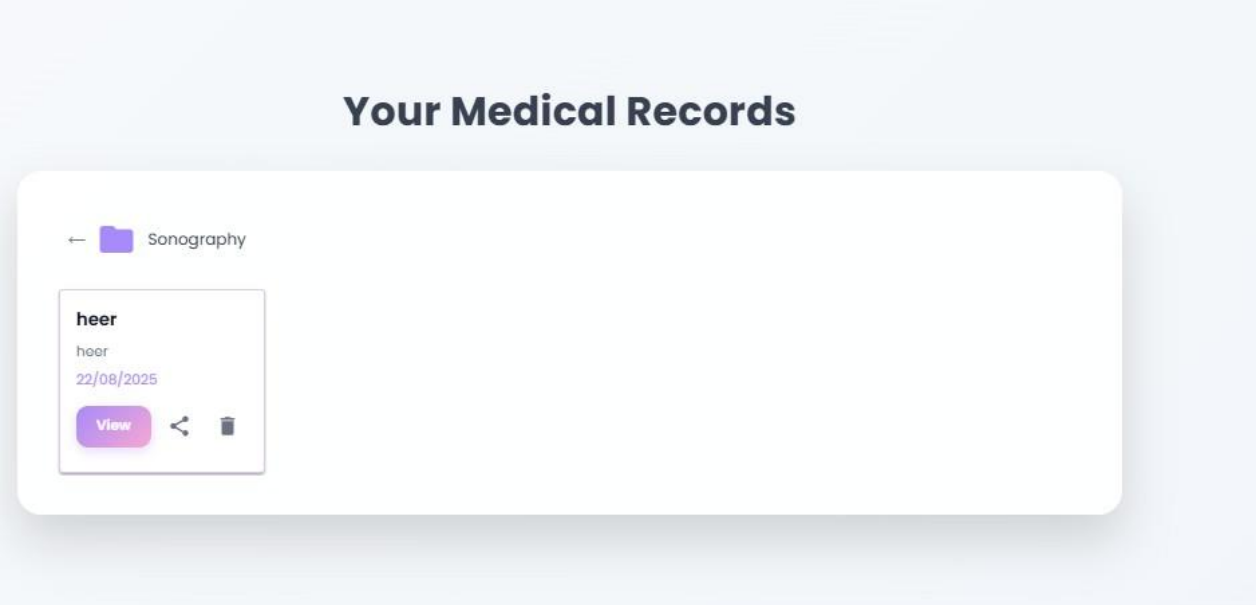


Fig 6.9 Medical Record

The screenshot displays the interface where patients can review and access their previously uploaded medical records.

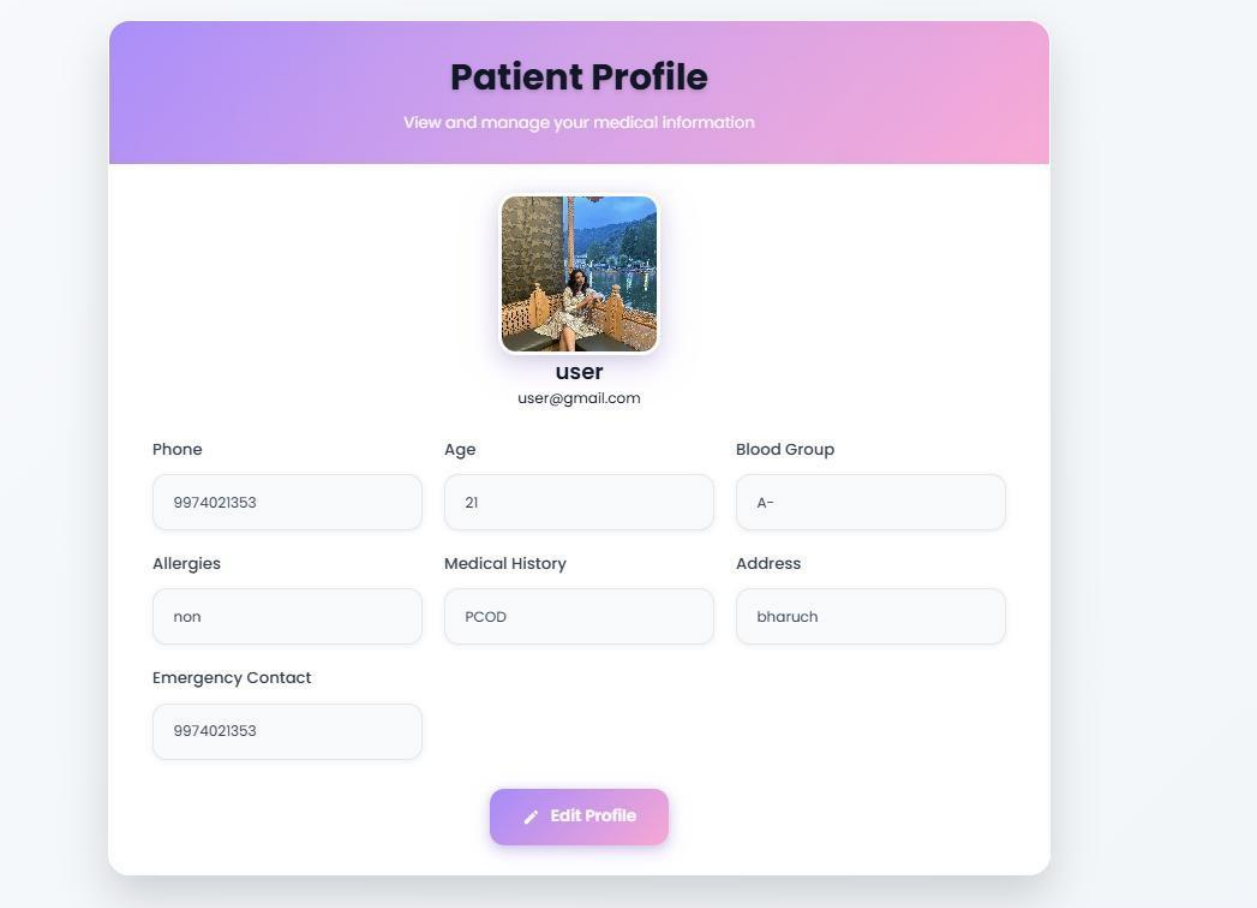


Fig 6.10 Patient Profile

The screenshot shows the patient profile page of the MediLog system, displaying personal details, medical history, and options to manage account information.

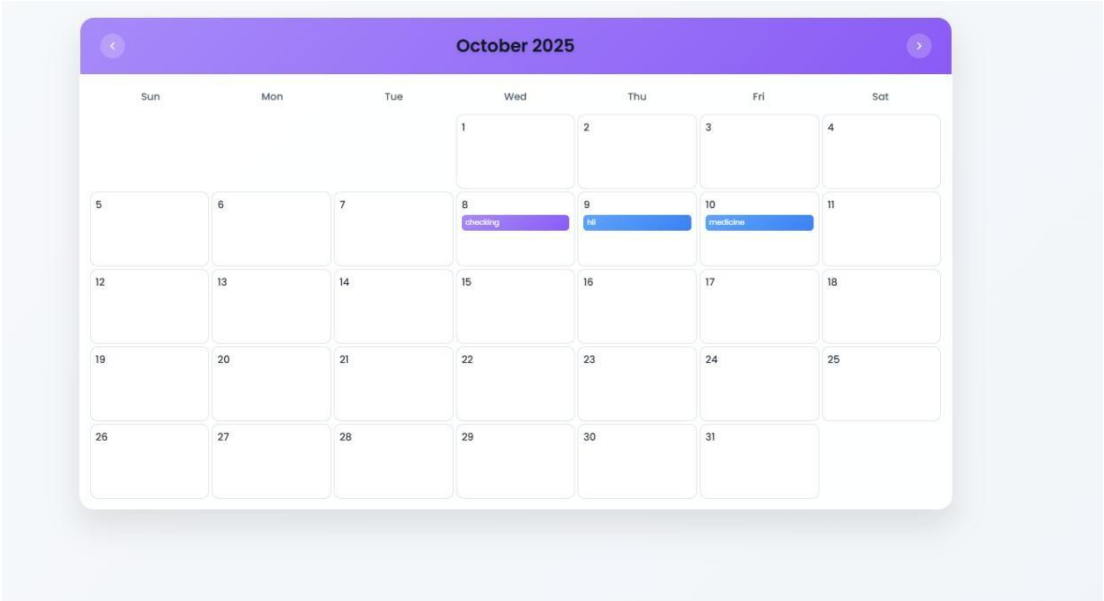


Fig 6.12 calendar

The screenshot displays the appointment scheduler page of the MediLog system, allowing patients to book, view, and manage their doctor appointments conveniently.

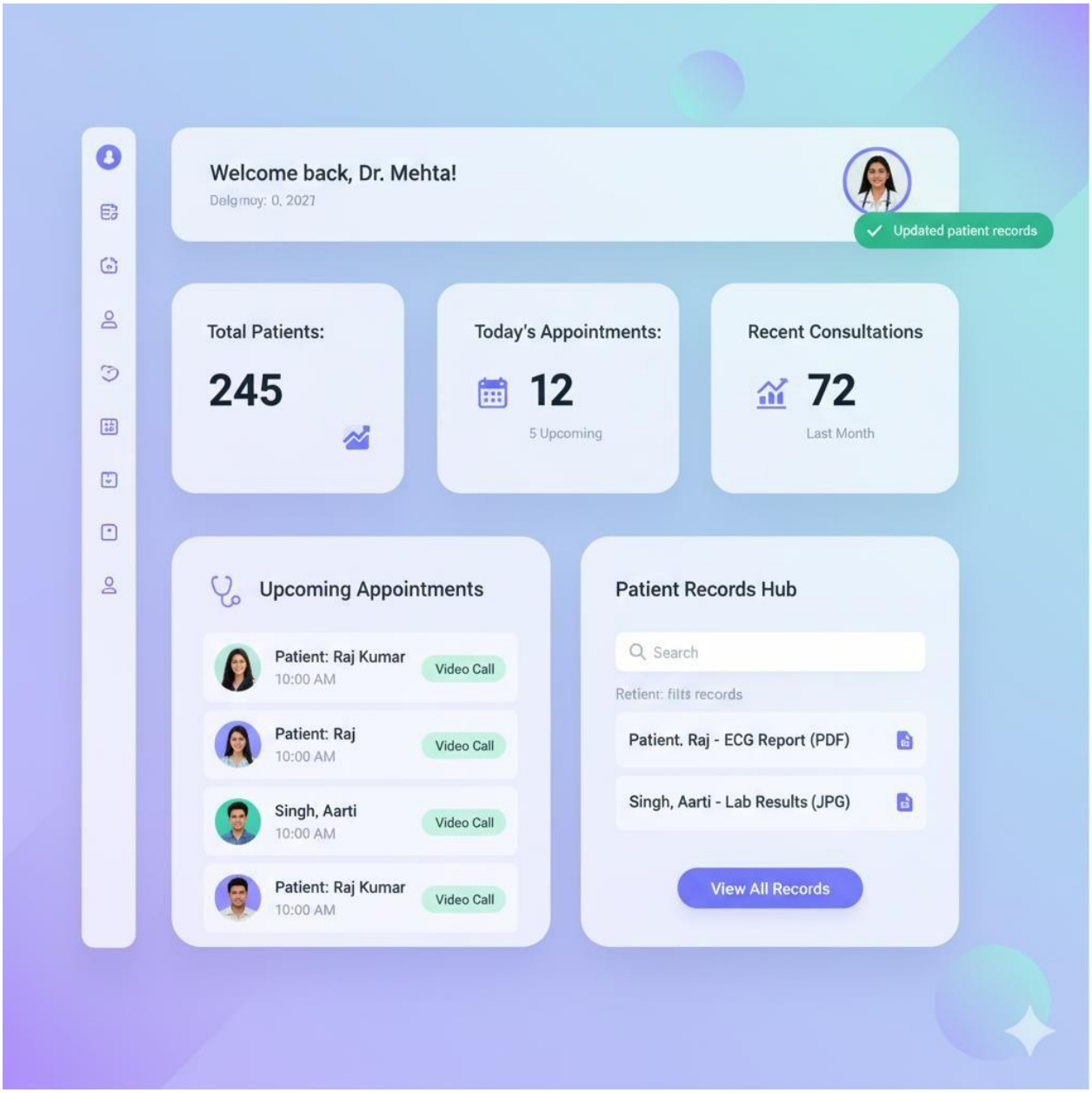


Fig 6.13 Doctor Dashboard

The screenshot shows the doctor dashboard of the MediLog system, providing doctors with access to patient records, appointments, and essential management tools.

CHAPTER – 7

Future Work

1. **Integration with Wearable Devices:**
   Enhance the model's performance by increasing the diversity of student data. This can include adding different types of user queries, feedback, and profiles from various educational institutions or student demographics to cover a broader range of counseling needs.
2. **AI-Based Health Insights:**
   Develop AI algorithms capable of analyzing uploaded medical records to detect early warning signs, provide insights, and suggest preventive health measures. This can enhance the system's usefulness by turning passive record storage into an intelligent health monitoring tool.
3. **Telemedicine Integration:**
   Introduce live video consultation and chat support for virtual doctor-patient interaction. This feature would allow patients to consult doctors remotely, reducing the need for physical visits and ensuring continued care accessibility.
4. **Multi-Language and Accessibility Support:**
   Expand the platform to support multiple regional languages and accessibility tools for differently-abled users. This will increase the portal's usability across diverse
5. **Automated Prescription Management:**
   Implement digital prescription generation and refill reminders to ensure patients follow prescribed treatments and medication schedules without missing doses.
6. **Data Analytics Dashboard:**
   Add analytics tools to allow doctors to visualize patient data trends over time, identify potential health issues, and make data-driven treatment decisions.
7. **Integration with Hospital Management Systems (HMS):**
   Connect MediLog with existing hospital databases and ERP systems to enable seamless information sharing between departments and improve operational efficiency.
8. **Enhanced Security through Blockchain:**
   Explore the use of blockchain technology to create a tamper-proof system for medical data storage and sharing, ensuring maximum data privacy, transparency, and trust.
9. **Smart Appointment Scheduling:**
   Develop an AI-based appointment scheduler that predicts optimal times for consultations and sends intelligent reminders to both patients and doctors to minimize missed appointments.
10. **Mobile Application Development:**
    Create a dedicated mobile application version of MediLog to provide real-time access to records, reminders, and communication on the go, enhancing portability and convenience for all users.

CHAPTER – 8

Limitations

1. **Limited Diagnostic Capability:**
   MediLog currently focuses on record management and does not include automated diagnosis or clinical decision support, which limits its use for advanced medical analysis.

2. **Dependence on Internet Connectivity:**
   As a web-based system, MediLog requires stable internet access. Users in rural or low-connectivity areas may face challenges accessing the portal effectively.

3. **Data Privacy and Compliance Issues:**
   Ensuring full compliance with medical data protection laws (like HIPAA or India's Personal Data Protection Act) is complex and requires continuous security monitoring.

4. **Limited Real-Time Monitoring:**
   The current system does not provide real-time patient health tracking or emergency alerts, which could be critical in urgent care scenarios.

5. **User Adoption Challenges:**
   Some patients or healthcare providers may be hesitant to shift from paper-based or legacy systems to digital platforms, slowing down overall adoption.

6. **Maintenance and Scalability:**
   Continuous updates, server maintenance, and database scaling are necessary to keep the system responsive and secure, which can require significant technical and financial resources.

7. **Potential Data Misuse:**
   If not properly monitored, unauthorized access or data sharing could lead to privacy concerns or misuse of sensitive medical information.

8. **Limited AI Integration:**
   Although the system includes AI-based reminders, it lacks comprehensive AI healthcare analytics, which could be developed to improve predictive and preventive capabilities.

CHAPTER – 9

Conclusion

The **MediLog Portal** represents a significant step toward the digital transformation of healthcare record management. It provides a secure, accessible, and user-friendly platform that bridges the gap between patients and doctors, simplifying the sharing and storage of vital medical information. By leveraging the power of the **MERN Stack** and **Cloudinary**, the system ensures scalability, data integrity, and smooth performance across devices. MediLog enables patients to take control of their health data while giving doctors the ability to monitor and assist patients more effectively through a structured digital environment.

While the project successfully achieves its goal of digitizing healthcare records and enabling efficient interaction between users, certain limitations such as internet dependency, maintenance overhead, and limited diagnostic capability remain. Future developments can focus on integrating advanced technologies like **AI-based analytics**, **wearable device connectivity**, and **telemedicine** to enhance its scope and usability.

In conclusion, MediLog lays a strong foundation for smarter and more connected healthcare management. With continuous improvement and technological innovation, it has the potential to evolve into a comprehensive digital health ecosystem that improves patient outcomes, optimizes doctor-patient collaboration, and contributes significantly to the modernization of healthcare infrastructure.

Chapter – 10

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